

MAD-1 Project Report

1. Student Details

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About me: I am a Student at IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Trekking Management App

Problem Statement:

To design and build a web-based application that allows **Admin, Trek Staff, and Users (Trekksers)** to interact with the system based on their roles and make managing trek approvals, tracking bookings, avoiding overbooking, and maintaining trek history easy.

Approach:

The app was built using Flask as a backend framework with a modular structure. It allows users(trekksers) to book treks and monitor their trekking and booking history, and allows admin and staff to manage treks and trekksers active on the app.

3. AI/LLM Declaration:

I used **Gemini** to assist in writing CSS styling and fixing minor bugs.

The extent of AI/LLM usage is around **15-20%** limited to code suggestions and documentation formatting.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
HTML5 & CSS3	For the custom Alpine Theme, Flexbox/Grid layouts, and custom UI cards.
SQLite	Lightweight local database for storing user data

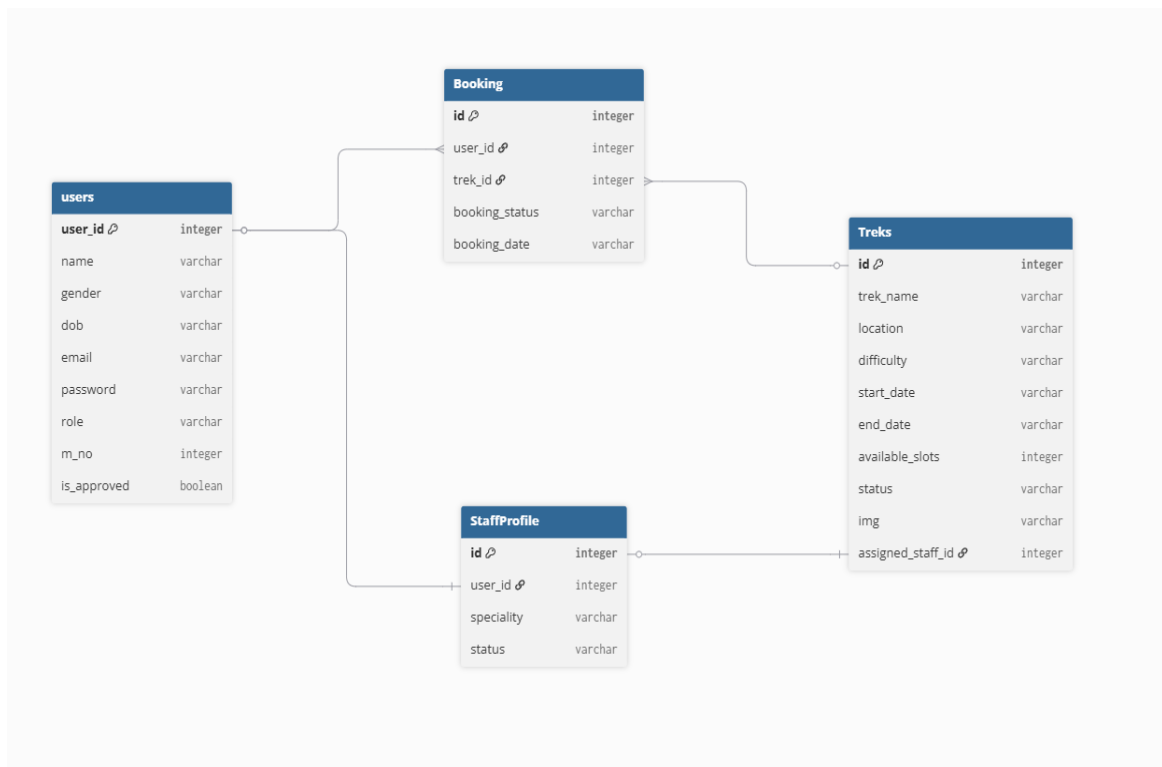
5. Database Schema / ER Diagram

Tables:

1. **User** — stores user profile details (*user_id, name, gender, dob, email, password, role, m_no, is_approved*)
2. **StaffProfile** — stores staff specific details (*id, user_id, speciality, status*)
3. **Treks** — stores Trek specific details (*id, trek_name, location, difficulty, start_date, end_date, available_slots, status, img, assigned_staff_id*)
4. **Booking** — stores bookig related details (*id, trek_id, user_id, booking_status, booking_date*)

Relationships:

- One-to-One → User → Staff Profile
- One-to-Many → User → Booking
- One-to-One → Trek Staff → Trek
- One-to-Many → Trek → Booking



7. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using SQLAlchemy
- **/templates** – Jinja2 HTML templates
- **/static** – CSS and images

Implemented Features:

Admin approves staff and adds trek to the app

Admin manages staff, treks and users with their respective booking

Staff can apply and after approval can login in the app

Staff can view and manage assigned treks with its participants

User can self-register and login in the app

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1EymkLPishtMeGGCzt9IEOs_gW4l1q-5F/view?usp=sharing